

Sentiment Analysis on IMDB MovieReviews

Overview

This project is a Natural Language Processing (NLP) and Deep Learning model that classifies movie reviews as Positive or Negative.

The model uses Word2Vec embeddings and a PyTorch Neural Network. Problem Statement The goal is to detect whether a movie review is positive or negative.

Workflow

1. Data Preprocessing

- Lowercasing text
- Removing HTML tags
- Removing punctuation
- Tokenization

2. Feature Extraction

- Pre-trained Google News Word2Vec (300-dimensional vectors)
- Averaging word vectors for sentence representation

3. Model Architecture

- Input Layer: 300 features
- Hidden Layer: 128 neurons (ReLU)
- Output Layer: 1 neuron (Sigmoid)

4. Training Details

- Loss Function: Binary Cross Entropy
- Optimizer: Adam
- Epochs: 200

Dataset

IMDB MovieReviews Dataset (50,000 reviews)

Evaluation

Test Accuracy: 82.41%

Technologies Used

- Python

- Pandas-

- NumPy-

- Scikit-learn –

- Gensim (Word2Vec)

- PyTorch Future Improvements

- Use BERT embeddings

Author

Youstina Eid